

TECHBIG SOLUTIONS

C & C++ Programming – Detailed Syllabus

Course Level: Beginner → Intermediate → Advanced

Recommended Duration: 120–180 Hours

Training Approach: Theory + Hands-on Practice + Projects + Interview Preparation

Course Objective

To provide a complete, industry-oriented learning path in C and modern C++, covering programming fundamentals, OOP, memory management, STL, data structures, algorithms, concurrency, development tools, projects and interview preparation.

MODULE 1 – Introduction to Programming

- Programming concepts and programming languages
- Compiler vs interpreter; source, object and executable code
- Algorithms, flowcharts and pseudocode
- Problem-solving and programming logic
- Introduction to C and C++; applications and C vs C++
- Development environment setup, compilation, execution and debugging

MODULE 2 – C Fundamentals

- Structure of a C program and main()
- Header files and preprocessor directives
- Keywords, identifiers, variables and constants
- Data types and type modifiers
- Type conversion and type casting
- printf(), scanf(), escape sequences
- Practical: Hello World, student details, calculator, temperature and area programs

MODULE 3 – Operators and Expressions

- Arithmetic, relational and logical operators
- Assignment and increment/decrement operators
- Conditional and bitwise operators
- Operator precedence and associativity
- Expressions and type casting

MODULE 4 – Decision Making

- if, if-else, nested if and else-if
- switch-case and nested switch
- Conditional operator
- Practical: largest number, grade calculator, electricity bill, ATM menu and result system

MODULE 5 – Loops and Iteration

- for, while and do-while loops
- Nested loops, break and continue
- Factorial, Fibonacci, prime, Armstrong, palindrome and reverse-number programs
- Number and star pattern programs

MODULE 6 – Functions in C

- Function declaration, definition and calling
- Parameters, arguments and return values
- Call by value and recursion
- Variable scope and storage classes
- auto, static, extern and register
- Menu-driven and recursive applications

MODULE 7 – Arrays

- One-dimensional, two-dimensional and multidimensional arrays
- Initialization and traversal
- Searching and sorting
- Linear search, binary search, bubble sort and selection sort
- Matrix addition, multiplication and transpose

MODULE 8 – Strings in C

- Character arrays and string input/output
- strlen(), strcpy(), strcat(), strcmp()
- String searching and manipulation
- Reverse string, palindrome, character and word counting

MODULE 9 – Pointers

- Pointer declaration, address operator and dereferencing
- Pointer arithmetic
- Pointers with arrays, strings and functions
- Pointer to pointer and function pointers
- malloc(), calloc(), realloc(), free()
- Memory leaks and dangling pointers

MODULE 10 – Structures, Unions and Enumerations

- Structures, nested structures and arrays of structures
- Structure pointers and functions
- Unions, enumerations and typedef
- Project: Student Management System

MODULE 11 – File Handling in C

- Files and streams; text and binary files
- fopen(), fclose(), fprintf(), fscanf(), fgets(), fputs()
- fread(), fwrite() and random access
- File error handling
- Project: Employee Record Management System

MODULE 12 – Advanced C Programming

- Preprocessor, macros and conditional compilation
- Header files and command-line arguments
- Bit manipulation and bit fields
- const and volatile
- Memory layout; stack vs heap
- Compilation, linking, debugging and common errors

MODULE 13 – Introduction to C++

- History of C++; C vs C++
- C++ compilation and program structure
- Namespaces and std
- cin, cout, cerr and clog
- References and nullptr

MODULE 14 – C++ Fundamentals

- Variables, constants, data types and operators
- Expressions and type conversion
- Scope, auto, const and constexpr
- enum class and nullptr

MODULE 15 – Object-Oriented Programming

- Classes and objects
- Data members and member functions
- public, private and protected
- Encapsulation and abstraction
- Inheritance and polymorphism
- Practical: employee, bank account, student and product classes

MODULE 16 – Constructors and Destructors

- Default, parameterized and copy constructors
- Move constructors
- Constructor overloading
- Destructors and initialization lists
- Delegating constructors and explicit constructors

MODULE 17 – Inheritance

- Single, multiple, multilevel and hierarchical inheritance
- Hybrid inheritance
- Constructor/destructor order
- Virtual inheritance and diamond problem

MODULE 18 – Polymorphism

- Function overloading and operator overloading
- Virtual functions and overriding
- Pure virtual functions and abstract classes
- Virtual destructors and dynamic binding

MODULE 19 – Operator Overloading

- Unary and binary operators
- Arithmetic and comparison operators
- Stream operators
- Assignment and increment/decrement operators
- Function-call and subscript operators

MODULE 20 – Templates

- Function and class templates
- Template parameters and type deduction
- Multiple template parameters
- Template specialization
- Variadic templates and introduction to concepts

MODULE 21 – Exception Handling

- try, catch and throw
- Multiple catch blocks and custom exceptions
- Standard exceptions
- Exception safety and RAII

MODULE 22 – STL (Standard Template Library)

- Containers: vector, array, deque, list, forward_list
- stack, queue, priority_queue, set, multiset, map, multimap
- unordered_set and unordered_map
- Iterators and iterator categories
- Algorithms: sort, find, count, reverse, binary_search, lower_bound, upper_bound, min_element, max_element, accumulate

MODULE 23 – Modern C++

- C++11: auto, range-for, lambdas, nullptr, smart pointers, move semantics, rvalue references, constexpr, enum class, threads
- C++14: generic lambdas, relaxed constexpr and variable templates

- C++17: structured bindings, if constexpr, optional, variant, any, string_view, filesystem and parallel algorithms
- C++20: concepts, ranges, coroutines, modules, spaceship operator, span, consteval and constexpr
- C++23: expected, print, generator, mdspan, flat_map, flat_set, stacktrace, to_underlying, byteswap, explicit object parameter and multidimensional subscript

MODULE 24 – Smart Pointers & Modern Memory Management

- Raw pointers vs smart pointers
- unique_ptr, shared_ptr and weak_ptr
- Ownership and reference counting
- Move semantics and RAI
- Custom deleters and memory-leak prevention

MODULE 25 – Lambda Expressions

- Lambda syntax and captures
- Capture by value and reference
- Generic and mutable lambdas
- Lambdas with STL algorithms
- Higher-order programming concepts

MODULE 26 – File and Directory Programming

- ifstream, ofstream and fstream
- Text and binary files
- Serialization concepts
- Filesystem library and directory operations
- File permissions and error handling

MODULE 27 – Data Structures Using C++

- Arrays and linked lists
- Doubly and circular linked lists
- Stacks, queues and deques
- Trees, binary trees and BST
- AVL trees, heaps and priority queues
- Graphs and traversal
- BFS, DFS, recursion, backtracking, dynamic programming and greedy algorithms

MODULE 28 – Searching and Sorting

- Linear and binary search
- Hash-based searching
- Bubble, selection and insertion sort
- Merge, quick and heap sort
- Counting and radix sort
- Big-O, Big-Ω and Big-Θ; time and space complexity

MODULE 29 – Multithreading and Concurrency

- Processes vs threads
- `std::thread`, `join` and `detach`
- `Mutex`, `lock_guard` and `unique_lock`
- Condition variables and atomic operations
- Futures, promises and `async`
- Race conditions, deadlocks and thread safety

MODULE 30 – Networking Fundamentals

- Networking concepts and TCP/IP basics
- Client-server architecture
- Sockets, TCP and UDP concepts
- Client/server programming
- Network error handling and introduction to HTTP communication

MODULE 31 – Database Connectivity

- Database concepts and SQL fundamentals
- Tables and CRUD operations
- C++ database connectivity
- Prepared statements and transactions
- Database error handling

MODULE 32 – Debugging & Development Tools

- Compiler, runtime and logical errors
- Breakpoints, watches and call stack
- Memory debugging and unit testing concepts
- Build systems and CMake introduction
- Git and GitHub basics

MODULE 33 – Software Development Practices

- Coding standards and naming conventions
- Clean and modular code
- Header/source separation and reusable components
- Documentation and version control
- Code review, testing, debugging and performance optimization

MODULE 34 – Interview Preparation

- C interview: pointers, memory, structures, arrays, strings, functions, recursion, files and preprocessor
- C++ interview: OOP, constructors/destructors, virtual functions, inheritance, polymorphism, templates, STL, smart pointers, move semantics, lambdas and multithreading
- 100+ coding problems, logical reasoning, data-structure exercises, algorithm questions and debugging tests

MODULE 35 – Industry Projects

- Beginner: Calculator, Student Mark Calculator, Number Guessing Game, Banking Calculator, Electricity Bill Generator
- Intermediate: Student Management, Employee Management, Library Management, Inventory Management, Hospital Management and Banking Management
- Advanced: Banking Application, Inventory System, Payroll System, Hotel Management, E-Commerce Backend, File Management, Client-Server and Multithreaded Applications

FINAL CAPSTONE PROJECT

- Develop a complete C++ industry application.
- Integrate OOP, STL, data structures and algorithms.
- Implement file/database handling and exception handling.
- Use smart pointers and modern C++ practices.
- Apply multithreading where appropriate.
- Use CMake, Git/GitHub, debugging and testing.
- Prepare technical documentation and project presentation.

TRAINING OUTCOMES

- Develop applications using C and C++.
- Apply object-oriented programming effectively.
- Work confidently with pointers and memory management.
- Use STL containers, iterators and algorithms.
- Implement data structures and algorithms.
- Build modern C++ applications.
- Work with files, databases and concurrency.
- Debug, test and optimize programs.
- Build portfolio-ready projects.
- Prepare for C/C++ developer interviews.

TECHBIG SOLUTIONS

Professional Training • Practical Learning • Industry-Oriented Skills

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